

Practical no. 16

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Class :- CO3K – B

Subject :- DSU

Program :-

```
#include<stdio.h>
#include<conio.h>
#define MAX 5
int stack[MAX];
int top = -1;
void push(int value){
if(top==MAX-1){
printf("\nStack Overflow!");
}
else{
top++;
stack[top]=value;
printf("\n%d pushed into stack.",value);
}
}
void pop(){
if(top==-1){
printf("\nStack empty");
}
else{
printf("\n%d popped from stack.",stack[top]);
top--;
}
}
void display(){
int i;
if(top==-1){
printf("\nStack is empty");
}
else{
printf("\nStack Elements are :- ");
for(i=top;i>=0;i--){
printf("\n%d",stack[i]);
}
printf("\n");
}
}
```

```
int main(){
int ch,value;
clrscr();
while(1){
printf("\n---STACK MENU---");
printf("\n1.Push\n2.Pop\n3.Display\n4.Exit");
printf("\nEnter your choice :- ");
scanf("%d",&ch);
switch(ch){
case 1:
printf("\nEnter value to push :- ");
scanf("%d",&value);
push(value);
break;
case 2:
pop();
break;
case 3:
display();
break;
case 4:
return 0;
default:
printf("\nInvalid Choice!");
}
}
getch();
return 0;
}
```

OUTPUT :-

```
DOSBox 0.74, Cpu speed: max 100%

---STACK MENU---
1.Push
2.Pop
3.Display
4.Exit
Enter your choice :- 1

Enter value to push :- 10

10 pushed into stack.

---STACK MENU---
1.Push
2.Pop
3.Display
4.Exit
Enter your choice :- 1

Enter value to push :- 20

20 pushed into stack.

---STACK MENU---
1.Push
2.Pop
3.Display
4.Exit
Enter your choice :- 3

Stack Elements are :-
20
10

---STACK MENU---
1.Push
2.Pop
3.Display
4.Exit
Enter your choice :- 2

20 popped from stack.
```